$t=1$

Tutoria-8 (Decision-Tree classif

ID3 → entropy and Information gain (IG)

(randomness)

Entropy (parent) : probability

↓ output

$P_{spam} = 5/10 = 0.5$

$P_{N.S} = 5/10 = 0.5$

$E \downarrow - IG \uparrow$

$E \uparrow \rightarrow IG \downarrow$

$$S = -\sum_{i=1}^{2(class)} P_i * \log_2(P_i)$$

$$= -[(0.5 \times \log_2(0.5)) + (0.5 \times \log_2(0.5))]$$

$$= -[-0.150 + (-0.150)]$$

$$= +[1]$$

entropy (money keywords)

0 — spam = 0, Nspam = 5 } $S_0 = 0$

1 — sp = 5, Nsp = 0 } $S_1 = 0$

$S_1 = 1\log_2(1) + 0$
$= 0$

$S_0 = -\sum_i p_i \times \log_2(p_i)$
$= 0 + 1\log_2(1)$
$= 0$

IG (Money keywords)

feature

$= \text{entropy(parent)} - \sum_K \frac{N_K}{N} \text{entrophy}(K)$

$= 1 - [\frac{1}{2} \times 0 + \frac{1}{2} \times 0] = 1$

⇒ Entrophy (Has Attach)

$\frac{1}{2} + \frac{N_K}{N}$

0 — S = 2, N = 3

1 — S = 3, N = 2

$S_0 = -[\frac{2}{5}\log_2(\frac{2}{5}) + \frac{3}{5}\log_2(\frac{3}{5})]$
$= 0.96$

$S_1 = -[\frac{3}{5}\log_2(\frac{3}{5}) + \frac{2}{5}\log_2[\frac{2}{5}]]$
$= 0.96$

IG1 (Has Attach)

$= \text{Ent(parent)} - \sum_K \frac{N_K}{N} \text{entropy}(K)$

$= 1 - [\frac{1}{2} \times 0.9 + \frac{1}{2} \times 0.96] = 0.04$

⇒ entropy (sender known) → entropy (ssw words)

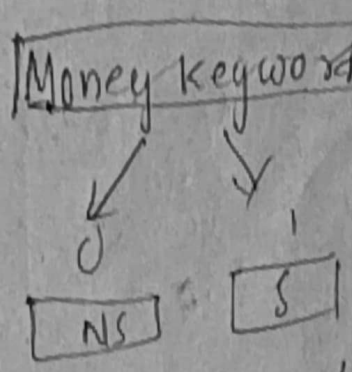
$S > 0$ - entrophy

0 — S = 4, N = 1 ⇒ $S_0 = 0$

1 — S = 1, N = 5 ⇒ $S_1 > 0$

IG < 1

IG(sender)
↓
< 1



Depth of the tree is 1

Two leaf nodes → NS, S

1/4/2026

Tutorial-10

S.No	F1 (Age)	F2 (Income)	F3 (Experience)	Target (Job offer)
S0	24	35	1	No -1
S1	30	50	3	Yes +1
S2	45	40	10	Yes +1
S3	35	30	7	No -1
S4	28	55	2	Yes +1

Target encoding

Yes = +1

No = -1

1) $w_i = \frac{1}{N} = \frac{1}{5} = 0.2$

2) Decision stump (weak learner) - Income (given info)

Income < 38 → No, ✗

≥ 38 → yes, ✓

N = 5

$\sum w_i I = [0 \times 0.2 + 0 \times 0.2 + 0 \times 0.2 + 0 \times 0.2 + 0 \times 0.2$